EMBEDDED ML · COMPUTER VISION · FLIGHT CONTROL

A 92 mm drone that follows one enrolled person.

The circuit board is the airframe. Flight firmware runs on an ESP32-S3; perception runs off-board on a laptop, over WiFi.

66 g · 292 tests · MIT licensed

Not yet flown. The boards are not fabricated.

Every number here is bench or simulation.

IDENTITY

Two signals, different jobs.

Face acquires and re-confirms the lock.

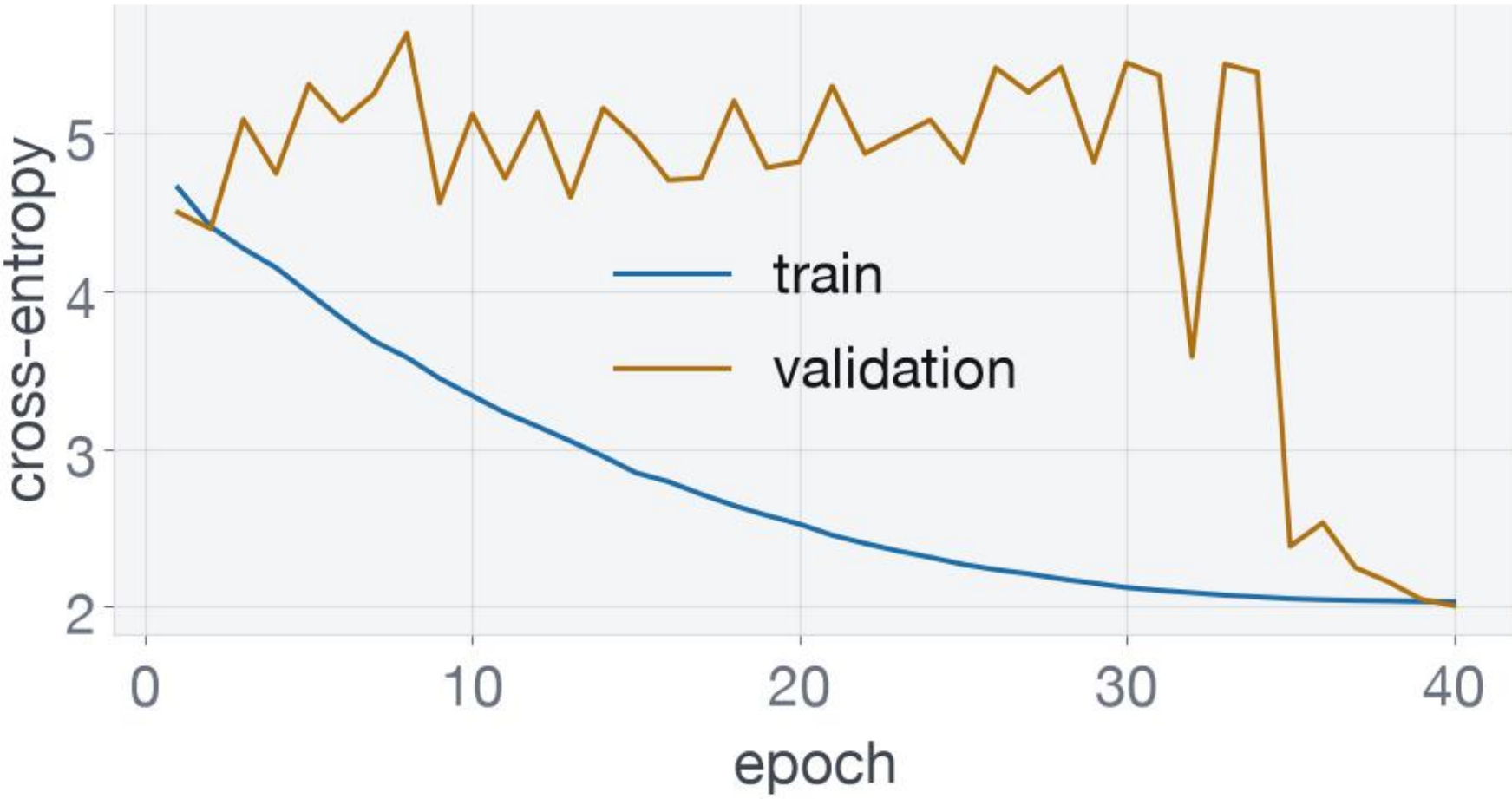
ArcFace 512-d, cosine threshold 0.36.

Body re-identification sustains it between face frames. 768-d embedding.

Only 26% of frames carry a usable face, so the body signal is doing most of the work.

A readable face scoring below 0.15 counts as evidence against, not as missing evidence.

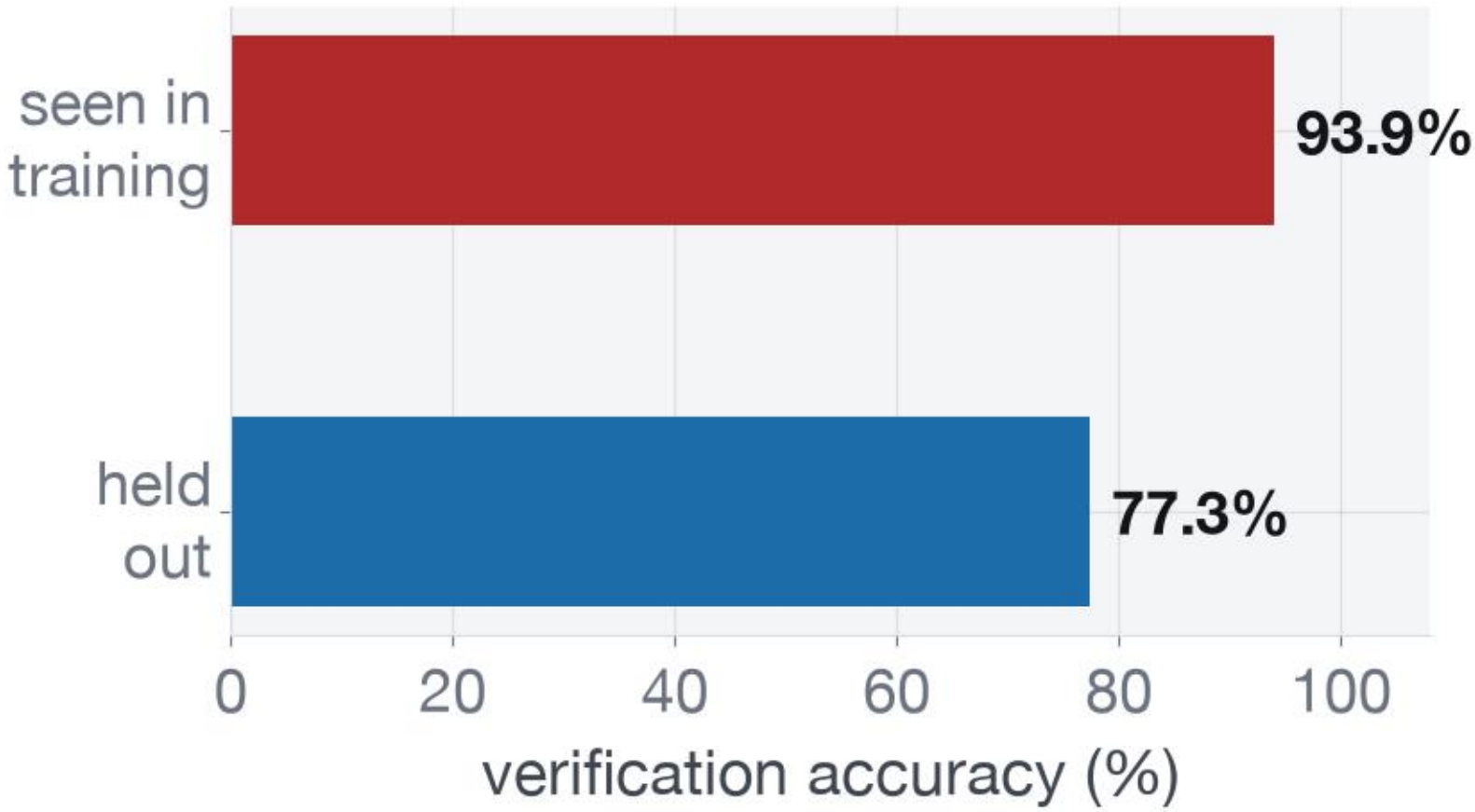
Trained an embedding on 4,324 photographs.



Labeled Faces in the Wild. 320,799 parameters, 1.283 MB. 40 epochs, 63s on an M4 GPU.

Validation loss climbs for 34 epochs, then drops as the cosine schedule decays the learning rate.

Split by identity, not by image.

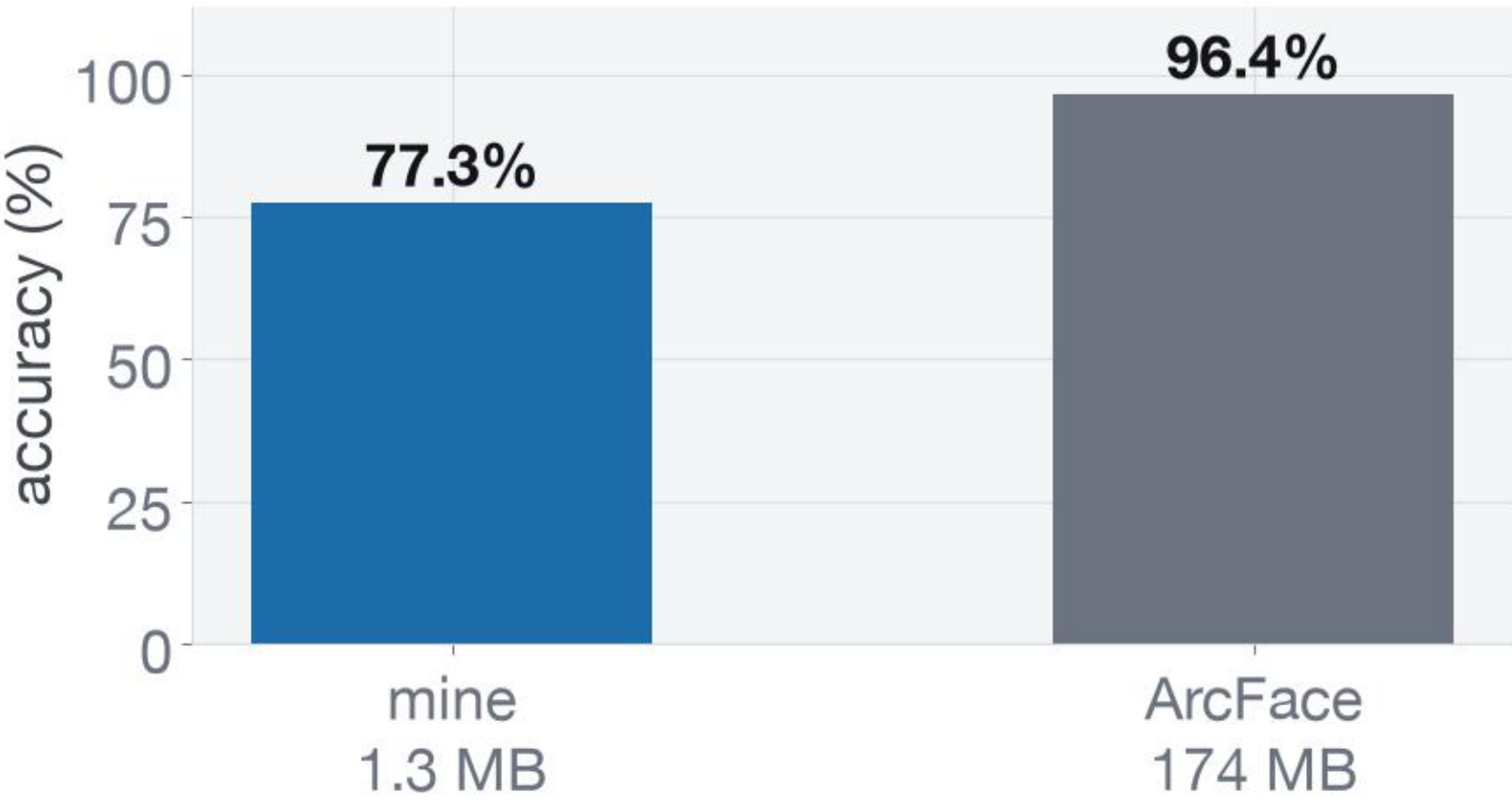


127 people to train on. 31 held out and never seen during training.

Splitting by image leaves the same faces on both sides. That is 16.6 points of inflation.

BENCHMARK

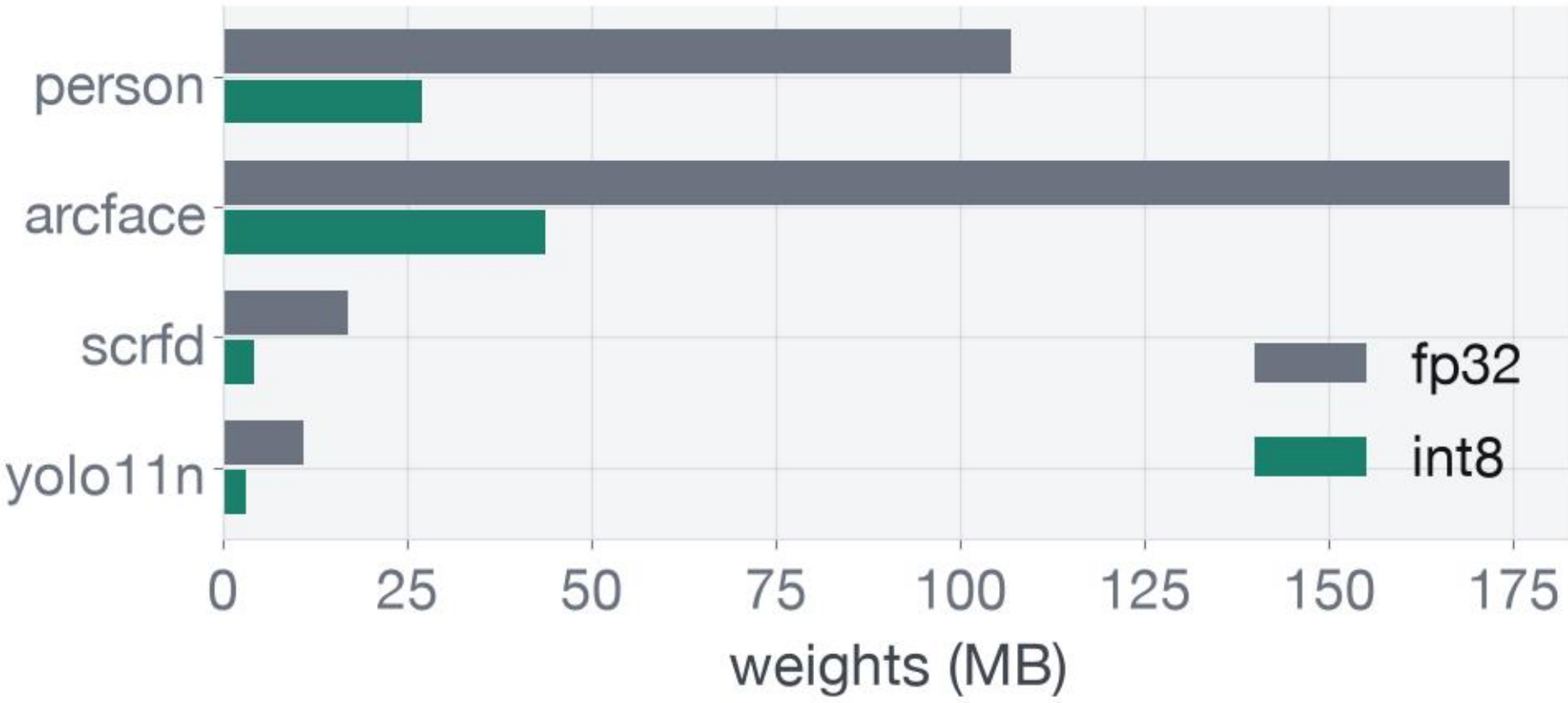
136x the weights buys 19 points.



Both scored on the same image pairs, from identities neither model was trained on.

ArcFace gets unaligned 62x47 crops here, which understates it. Its published LFW figure is 99.8%.

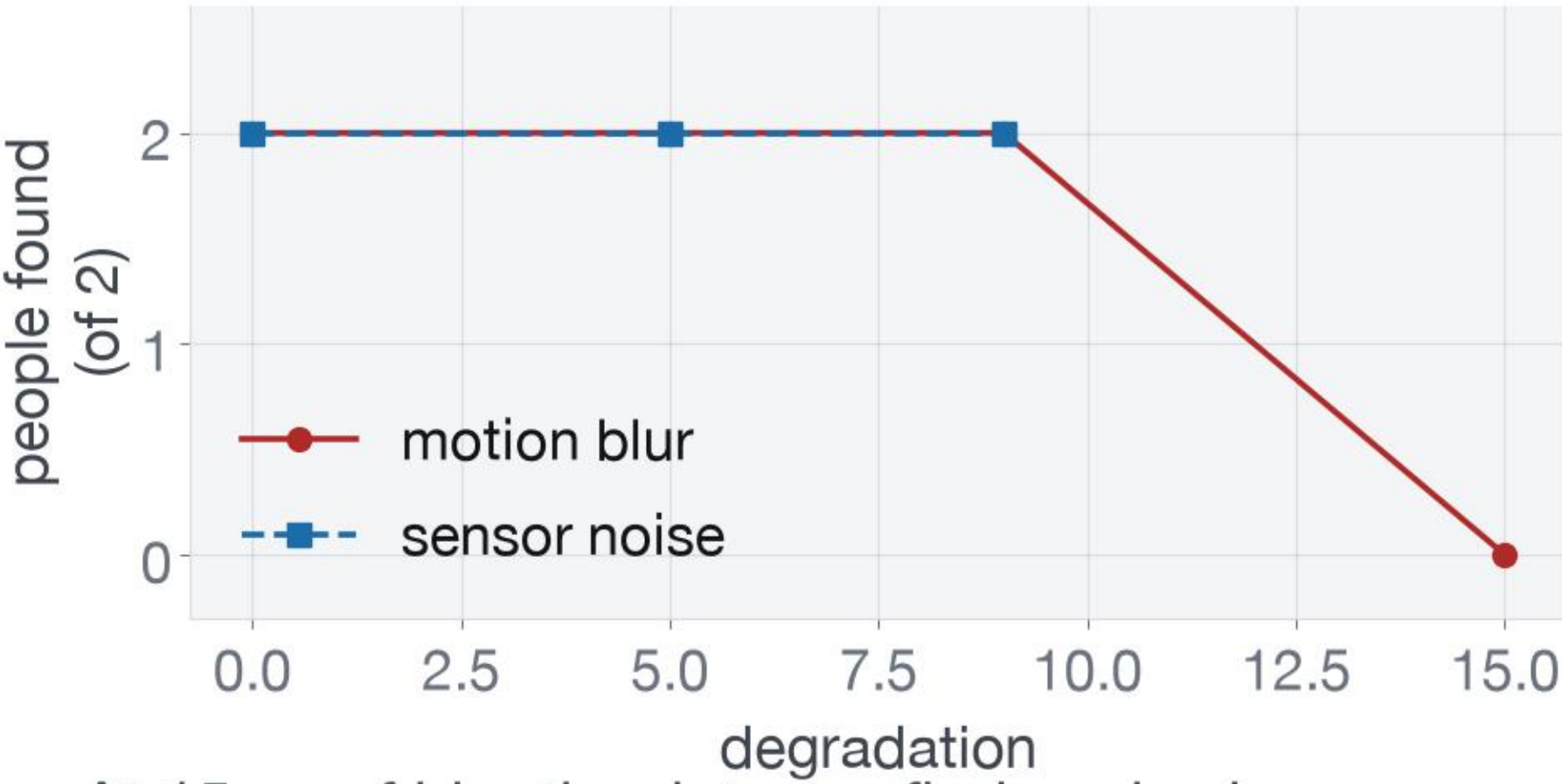
INT8 made it 4x smaller and 11% slower.



CoreML runs fp16 on the Neural Engine faster than the CPU runs INT8. It would win on a Raspberry Pi.

Blur kills detection.

Noise does not.

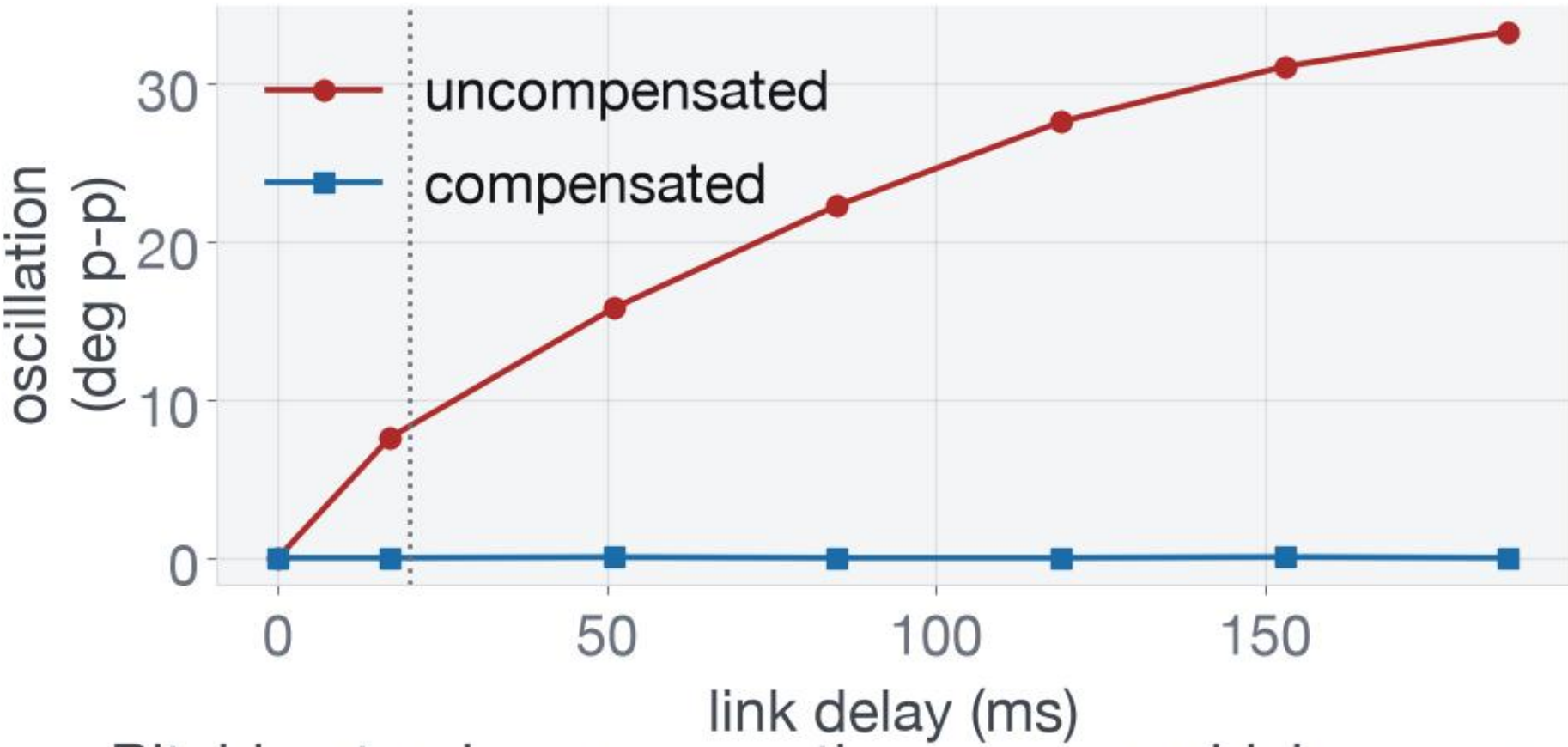


At 15 px of blur the detector finds nobody.

At 4x gain it still finds both people.

So the camera is set for short exposure and accepts the sensor noise that comes with it.

The vision loop closes through the airframe.



Pitching to chase moves the camera, which moves the subject in frame. Gain 34/26 = 1.31.

Above 1, the correction overshoots. Uncompensated it rings at one frame; the real RTT is 20 ms.

LIMITATIONS

What is not done.

The aircraft has never flown and the boards are not fabricated.

PID gains are scaled from typical values, not tuned on this airframe.

The stability result is simulated, with estimated mass properties.

The two-person test has never been run on two real people.

Listed because the numbers on the other slides are only worth anything if this slide exists.

Built end to end.

Board	generated from a Python netlist, with a release check before fab
Firmware	ESP32-S3 from the IMU registers up
Vision	ONNX Runtime and CoreML, no torch
Protocol	generated from one source, so the two ends cannot drift

207 Python and 85 native C++ tests

Code, write-up and every measurement:

github.com/Pai-88/pursuit-drone